COMPUTER NETWORKS (DU # 2304CS301)

UNIT 2: TRANSMISSION MEDIA

COMPLETE EXAM-READY STUDY GUIDE

Key Metadata	Details
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> **Exam Success Mandate**
>
> *This study guide covers 100% of the concepts, classifications, cable categories, wave propagation modes, connectors, and Bluetooth architectures from Unit 2. It is engineered to serve as your single, standalone resource for top exam marks.*

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CHAPTER 1

INTRODUCTION TO TRANSMISSION MEDIA & DESIGN FACTORS

1.1 What is Transmission Media?

Transmission media is the physical communication channel or path through which data, voice, or video signals are transmitted from a sender (Source) to a receiver (Destination) within a computer network. It resides at Layer 1 (Physical Layer) of the OSI model.

1.2 Broad Classification of Transmission Media

Transmission media is broadly divided into two major categories depending on whether physical solid conductors guide the signals or waves radiate freely through space:

Media Category	Synonym	Physical Conduit	Primary Examples
Wired Media	Guided / Bounded Media	Solid physical conductors (copper, glass fiber)	Twisted-Pair, Coaxial Cable, Fiber Optic
Wireless Media	Unguided / Unbounded Media	Air, vacuum, or free space (EM waves)	Radio Waves, Microwaves, Infrared, Bluetooth

1.3 Factors Affecting the Design of Transmission Media

When designing or selecting a transmission medium for a network, engineers evaluate key physical and operational characteristics:

1. Bandwidth: The maximum volume of data that can be transmitted across a communication channel in one second. Measured in bits per second (bps), Kbps, Mbps, or Gbps. Higher bandwidth enables faster data transfer rates.

2. Transmission Impairments: Signals undergo physical degradation during propagation, causing the received signal to differ from the original. The three primary impairments are:

• **Attenuation (Signal Energy Loss):** The reduction of signal strength over distance. Amplifiers or repeaters are installed along the channel to boost attenuated signals.

• **Distortion (Signal Shape Alteration):** Changes in signal shape occurring in composite signals because different frequency components travel at varying speeds.

• **Noise (Unwanted External Signals):** External electrical energy inserted into the signal (e.g., thermal noise, crosstalk, electromagnetic interference from motors or high-voltage lines).

> **Important Exam Point**
> *Transmission impairments destroy signal quality. Attenuation dictates the maximum distance between repeaters, while crosstalk and EMI govern the choice between UTP and STP cables.*

CHAPTER 2

GUIDED TRANSMISSION MEDIA (WIRED)

2.1 Twisted-Pair Cable

Twisted-pair cable consists of two separately insulated copper wires twisted together in a helical spiral. Twisting cancels out **Electromagnetic Interference (EMI)** and **crosstalk** from adjacent wire pairs. Voice phone lines operate at 0 to 3.5 kHz, while modern Ethernet twisted pairs support frequencies into the MHz and GHz spectrums.

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Types of Twisted-Pair Cables:

Cable Type	Shielding Mechanism	Key Advantages	Disadvantages / Limits
Unshielded Twisted Pair (UTP)	No metallic shield; outer plastic jacket only.	Low cost, lightweight, highly flexible, easy installation.	Prone to crosstalk & EMI; short distance due to attenuation.
Shielded Twisted Pair (STP)	Each pair/cable wrapped in metallic foil/braid.	Superior noise immunity, lower attenuation, higher data rates.	Higher cost, rigid, difficult installation requiring proper grounding.

UTP Cable Categories (Cat 1 to Cat 7):

Category	Max Data Speed / Bandwidth	Primary Application
Cat 1	Low speed (< 1 Mbps)	Traditional telephone voice lines
Cat 2	Up to 4 Mbps	Legacy PBX and token ring networks
Cat 3	Up to 10 Mbps (16 MHz)	10BASE-T Ethernet LANs
Cat 4	Up to 16 Mbps	Token Ring networks
Cat 5	Up to 100 Mbps (100 MHz)	100BASE-TX Fast Ethernet
Cat 5e	Up to 1 Gbps (100 MHz)	1000BASE-T Gigabit Ethernet (Enhanced crosstalk control)
Cat 6	Up to 10 Gbps (250 MHz)	10GBASE-T High-speed LANs (up to 55 meters)
Cat 7	Up to 10 Gbps+ (600 MHz)	Data centers & fully shielded high-frequency networks

Twisted-Pair Connectors: Standard Ethernet twisted-pair cables use the **RJ-45 (Registered Jack-45)** connector featuring an 8-pin conductor layout in male (plug) and female (jack) options.

2.2 Coaxial Cable

Coaxial cable carries high-frequency signals with minimal interference due to its 4 concentric layers: (1) **Inner Copper Conductor** carrying data, (2) **Dielectric Insulator**, (3) **Outer Braided Shield** blocking EMI, and (4) **Outer Protective Plastic Jacket**.

Coaxial Transmission Modes:

• **Baseband Transmission:** Transmits a single digital signal using the *entire cable bandwidth* (e.g., Analog CCTV camera).

• **Broadband Transmission:** Transmits *multiple analog signals simultaneously* by dividing bandwidth into distinct frequency channels using Frequency Division Multiplexing (FDM) (e.g., Cable TV carrying 100+ channels).

Coaxial Cable Connectors:

Connector Name	Description & Exam Function
BNC Connector	Bayonet Network Connector; locks RG-58/RG-62 coaxial cables to NICs & CCTV equipment.
BNC T-Connector	T-shaped connector with 3 ports; connects a computer workstation to the main coaxial bus cable.
BNC Terminator	50-Ohm resistive end cap placed at both physical ends of a bus topology to absorb signals & prevent reflection.
Barrel Connector	Inline female-to-female connector used to join two coaxial cables to extend total reach.

2.3 Fiber Optic Cable

Fiber optic cable transmits data as pulses of light through ultra-pure glass or plastic strands based on **Total Internal Reflection**.

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Layer	Material	Functional Purpose
1. Core	Glass / Plastic	Central light-guiding tube carrying optical signals.
2. Cladding	Glass / Plastic (Lower Refractive Index)	Reflects light pulses back into the core via Total Internal Reflection.
3. Coating	Buffer Plastic Coating	Protects fragile glass fiber against moisture, scratches, and damage.
4. Strength Member	Kevlar Synthetic Strands	Provides tensile strength to prevent stretching/bending during installation.
5. Outer Jacket	Durable Plastic Shield	Protects cable against heat, water, chemicals, and crushing forces.

Fiber Optic Connectors:

- **SC (Subscriber / Standard Connector):** Push-pull locking system; widely used in Cable TV.
- **ST (Straight Tip Connector):** Bayonet twist-lock mount; standard for network equipment.
- **MT-RJ (Mechanical Transfer-Registered Jack):** Compact dual-fiber connector matching RJ-45 size.

2.4 Master Comparison: Guided Transmission Media

Feature / Criteria	Twisted-Pair Cable	Coaxial Cable	Fiber Optic Cable
Signal Type	Electrical Signals	Electrical Signals	Light Signals (Photons)
Data Rate / Speed	Moderate (10 Mbps - 10 Gbps)	High (10 Mbps - 100 Mbps+)	Ultra High (100 Gbps+)
Bandwidth	Low to Moderate	Moderate to High	Extremely High
Attenuation	High (Short distance ~100m)	Moderate	Very Low (Tens of kilometers)
EMI Immunity	Low (UTP) to Moderate (STP)	High (Braided Shield)	100% Immune (Immune to EMI/RFI)
Cost	Lowest / Inexpensive	Moderate	High
Installation Ease	Very Easy	Moderate	Difficult (Requires splicing)
Flexibility & Weight	Lightweight & Flexible	Heavier & Stiff	Lightweight, Compact, Fragile

CHAPTER 3
UNGUIDED TRANSMISSION MEDIA (WIRELESS) & WAVE PROPAGATION

3.1 What is Unguided Media?

Unguided media (wireless communication) transmits electromagnetic waves through air, vacuum, or free space without using solid conductors. Signals radiate from antennas into open space.

3.2 Wireless Wave Propagation Modes

Propagation Mode	Frequency Range	Physical Mechanism	Primary Applications
Ground Propagation (Surface Wave)	< 2 MHz	Low-frequency waves hug Earth's surface and follow its curvature.	AM Radio, Submarine/Marine navigation.
Sky Propagation (Sky Wave)	2 MHz to 30 MHz	Waves shoot upward and reflect off the ionosphere back to Earth.	Shortwave radio, international broadcasting.
Line-of-Sight (LOS) Propagation	> 30 MHz	High-frequency waves travel straight between visible antennas.	FM Radio, TV, Satellite, Microwaves.

3.3 Radio Waves (3 kHz to 1 GHz)

Radio waves are omnidirectional electromagnetic waves that radiate in all directions from the antenna. Alignment is not required. They easily penetrate building walls and cover wide geographical areas (WANs).

3.4 Microwaves (1 GHz to 300 GHz)

Microwaves are high-frequency unidirectional waves traveling in narrow straight beams. They *cannot penetrate solid obstacles*, requiring strict Line-of-Sight (LOS) alignment.

Types of Microwave Systems:

• **Terrestrial Microwave (4 GHz - 23 GHz):** Ground-based parabolic dish or horn antennas on towers. Data rates reach 1-10 Mbps. Attenuates during heavy rain/fog.

• **Satellite Microwave (1 GHz - 40 GHz):** Space relay satellite. Ground terminal sends an **Uplink** signal to satellite, which amplifies and retransmits a **Downlink** signal back to Earth, covering global regions.

3.5 Infrared Waves (300 GHz to 400 THz)

Infrared (IR) waves are ultra-high frequency short-range signals. IR *cannot penetrate solid walls*, confining signals strictly to a single enclosed room for secure short-range control (e.g., TV remotes, wireless mice).

3.6 Master Comparison: Unguided Wireless Media

Feature	Radio Waves	Microwaves	Infrared Waves
Frequency Range	3 kHz – 1 GHz	1 GHz – 300 GHz	300 GHz – 400 THz
Directionality	Omnidirectional (All directions)	Unidirectional (Narrow Beam)	Unidirectional (Line-of-Sight)
Wall Penetration	Yes (Penetrates walls easily)	No (Blocked by obstacles)	No (Confined to single room)
Antenna Alignment	Not Required	Strictly Required (LOS)	Strictly Required (LOS)
Primary Uses	AM/FM Radio, TV Broadcast	Satellite, Cellular, Wi-Fi	TV Remote, Wireless Mouse

CHAPTER 4

BLUETOOTH TECHNOLOGY & WIRELESS PAN

4.1 Overview of Bluetooth Technology

Bluetooth is a low-power, short-range wireless Personal Area Network (WPAN) technology operating in the globally unlicensed **ISM band** (**2.4 GHz to 2.4835 GHz**) up to 10 meters at 720 Kbps to 3 Mbps.

Frequency-Hopping Spread Spectrum (FHSS): To prevent radio interference, Bluetooth hops across 79 channels 1,600 times per second.

4.2 Bluetooth Architectures: Piconet & Scatternet

1. Piconet Architecture: Consists of **1 Master** and up to **7 active Slaves** (max 8 active nodes). Up to 255 secondary devices can stay in a *parked state* (2^8 - 1 = 255). Direct slave-to-slave communication is forbidden — all data flows through the Master.

2. Scatternet Architecture: Formed by interconnecting 2+ Piconets via **Bridge Nodes** that relay traffic between cells. A node cannot be Master in two Piconets simultaneously.

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Piconet vs. Scatternet Comparison Matrix:

Feature / Dimension	Piconet Architecture	Scatternet Architecture
Network Structure	Single standalone Bluetooth cell.	Multi-cell network connecting 2+ Piconets.
Active Device Capacity	Maximum 8 nodes (1 Master + 7 Slaves).	Exceeds 8 nodes across interconnected cells.
Parked Device Limit	Up to 255 parked/paired devices (2^8 - 1).	Multiple parked devices per constituent Piconet.
Bridge Node Role	Not Applicable (No bridge required).	Bridge nodes relay data between adjacent Piconets.
Coverage Area	Small (~10 meters radius).	Extended coverage area via bridging.
Practical Example	Smartphone connected to smartwatch & earbuds.	Multiple laptops & peripherals sharing data in an office.

CHAPTER 5
QUESTION BANK & MODEL EXAM ANSWERS

Q1: Define Transmission Media and explain its types with a detailed classification diagram.
Answer Outline: Transmission media is the Physical Layer communication path that carries signals from Source to Destination. It is classified into **Guided (Wired)** media (Twisted-Pair, Coaxial, Fiber Optic) and **Unguided (Wireless)** media (Radio Waves, Microwaves, Infrared, Bluetooth). Factors affecting selection include Bandwidth, Transmission Impairments (Attenuation, Distortion, Noise), and cost.

Q2: Compare Shielded Twisted Pair (STP) and Unshielded Twisted Pair (UTP) cables.
Answer Outline: UTP lacks metal shielding, making it cheap, flexible, and easy to install, but prone to EMI and crosstalk (used in Cat 5e/Cat 6 LANs). STP wraps wire pairs in metallic foil, providing high noise immunity and lower attenuation, but is expensive, rigid, and requires proper grounding.

Q3: Explain the structure of a Fiber Optic Cable and why it offers superior performance.
Answer Outline: Fiber optic cable consists of 5 layers: Core (glass light tube), Cladding (reflects light via Total Internal Reflection), Coating (protective plastic), Strength Member (Kevlar), and Outer Jacket. It offers ultra-high bandwidth, long-distance transmission, low attenuation, and 100% immunity to Electromagnetic Interference (EMI).

Q4: Differentiate between Terrestrial Microwave and Satellite Microwave transmission.
Answer Outline: Terrestrial microwave uses ground-based towers (4-23 GHz) requiring clear line-of-sight over short distances. Satellite microwave uses orbiting space satellites (1-40 GHz) with Uplink and Downlink frequencies to amplify and relay signals across oceans and remote global regions.

Q5: Explain Bluetooth Architecture (Piconet vs Scatternet) in detail.
Answer Outline: A Piconet comprises 1 Master and up to 7 active Slaves (max 8 nodes) where slave-to-slave communication is prohibited. A Scatternet links multiple Piconets using Bridge Nodes that belong to both Piconets, expanding coverage and node count.

CHAPTER 6
MULTIPLE CHOICE QUESTIONS (MCQS) WITH EXPLANATIONS

1. Which transmission medium operates on the principle of Total Internal Reflection?

A) Coaxial Cable

B) Twisted-Pair Cable

C) Fiber Optic Cable

D) Infrared Waves

Correct Answer: C — *Explanation: Fiber optic core-cladding interface relies on Total Internal Reflection to keep light contained within the core.*

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2. What is the maximum number of active devices in a single Bluetooth Piconet?

A) 7

B) 8

C) 255

D) 16

Correct Answer: B — *Explanation: A Piconet contains 1 Master node and up to 7 active Slave nodes, making a maximum of 8 active devices.*

3. Which connector is specifically used to terminate coaxial cables in bus topologies to absorb signals?

A) RJ-45 Connector

B) BNC Terminator

C) SC Connector

D) Barrel Connector

Correct Answer: B — *Explanation: 50-Ohm BNC terminators are installed at both ends of a bus cable to prevent signal reflection.*

4. Which wave propagation mode reflects radio signals off the ionosphere for long-distance communication?

A) Ground Propagation

B) Sky Propagation

C) Line-of-Sight Propagation

D) Surface Propagation

Correct Answer: B — *Explanation: Sky propagation (2-30 MHz) bounces electromagnetic waves off the charged ionosphere back to Earth.*

5. Infrared communication is limited to short distances and single rooms because:

A) It requires a satellite

B) It cannot penetrate solid walls

C) It uses FHSS

D) It suffers high EMI

Correct Answer: B — *Explanation: Infrared signals cannot pass through solid walls or obstacles, confining signals to enclosed rooms.*

6. The frequency range of the Bluetooth ISM band is:

A) 3 kHz to 1 GHz

B) 1 GHz to 40 GHz

C) 2.4 GHz to 2.4835 GHz

D) 300 GHz to 400 THz

Correct Answer: C — *Explanation: Bluetooth operates in the unlicensed 2.4 GHz ISM band (2.400 - 2.4835 GHz).*

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CHAPTER 7
FINAL REVISION SHEET & CHEAT SHEET

Master Quick-Reference Tables:

1. Cable Category Quick Summary:

Category	Speed	Frequency	Common Use
Cat 3	10 Mbps	16 MHz	10BASE-T LAN
Cat 5e	1 Gbps	100 MHz	Gigabit LAN
Cat 6	10 Gbps	250 MHz	10G LAN (55m)
Cat 7	10 Gbps+	600 MHz	Data Centers

2. Electromagnetic Wave Spectrum Summary:

Wave Type	Frequency Spectrum	Directionality	Wall Penetration
Radio Waves	3 kHz – 1 GHz	Omnidirectional	Yes
Microwaves	1 GHz – 300 GHz	Unidirectional (LOS)	No
Infrared	300 GHz – 400 THz	Unidirectional (LOS)	No (Room-bound)

3. Connector Reference:

Medium	Primary Connectors Used
Twisted Pair	RJ-45 (Male plug / Female jack)
Coaxial Cable	BNC, BNC-T, BNC Terminator, Barrel Connector
Fiber Optic	SC, ST, MT-RJ

> **Final Exam Checklist for Tanish Uchhadiya**
> • *Remember that Fiber Optic cladding has a LOWER refractive index than the core.*
> • *A Bluetooth Piconet has EXACTLY 1 Master and MAX 7 active Slaves (8 total active nodes).*
> • *BNC terminators (50 ohms) are mandatory at both ends of a coaxial bus topology.*
> • *Sky propagation reflects off the Ionosphere (2-30 MHz); Ground propagation follows Earth's curve (<2 MHz).*